

Chandra Earl
Biodiversity Informatician
The Sampling Event Horizon: ***Lessons from the Publishing Frontier***
GBIF-NA Webinar 2026
October 22, 2026

Operated by Battelle

The Sampling Event Horizon: Lessons from the Publishing Frontier

A data provider's perspective from the National Ecological Observatory Network

© 2026 Chandra Earl. Licensed under CC BY 4.0 – https://creativecommons.org/licenses/by/4.0/

This material is based upon work supported by NSF's National Ecological Observatory Network which is a major facility fully funded by the U.S. National Science Foundation

PEOPLE

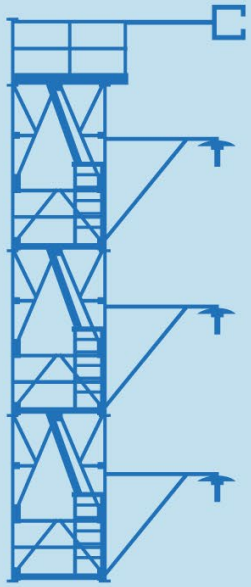
~450 total staff

300+ full time

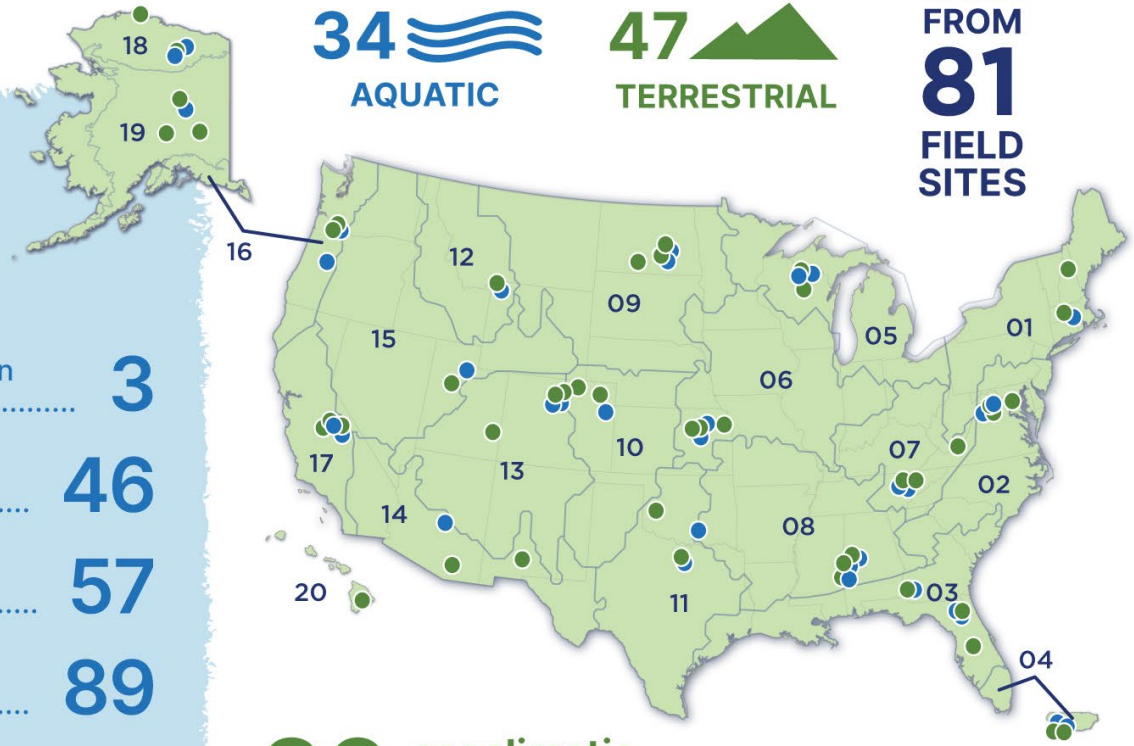
~150+ seasonal Domain techs

WHAT IS THE National Ecological Observatory Network?

PHYSICAL INFRASTRUCTURE



airborne observation platforms	3
flux towers	46
water quality stations	57
meteorological stations	89
groundwater wells	196
soil sensor arrays	230



34 AQUATIC

47 TERRESTRIAL

DATA FROM 81 FIELD SITES

20 ecoclimatic Domains

24 states +1 territory

NSF | neon Operated by Battelle

US NSF major facility operated by Battelle

research infrastructure

continental-scale observatory designed to enable understanding of US ecosystems

provider of Open & FAIR Data, Samples, & Infrastructure for Open Science

NEON's Linked Collection Systems

Standardized, collocated methods across 81 sites



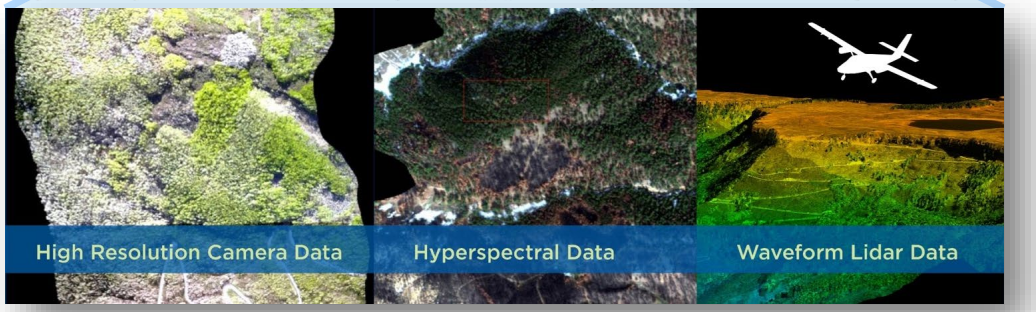
Automated Instruments



Airborne Remote Sensing



Observational Sampling



DATA PRODUCTS 180+



ASU Arizona State University Biorepository

100,000+ SAMPLES ADDED PER YEAR

© 2026 Chandra Earl. Licensed under CC BY 4.0 – https://creativecommons.org/licenses/by/4.0/

NSF neon Operated by Battelle

GBIF | Get data | How-to | Tools | Community | About | sunray1

Publisher | Since 20 September 2019

National Ecological Observatory Network

www.neonscience.org

1,452,530 occurrences | 47 published datasets | 1,002 citations

About | Metrics

Description

The National Science Foundation's National Ecological Observatory Network (NEON) is a continental-scale ecological observation facility, fully funded by NSF and operated by Battelle. NEON collects and provides open data from 81 field sites across the United States that characterize and quantify how our nation's ecosystems are changing. Each year, NEON collects and archives over 100,000 specimens and samples that complement the field observations and automated measurements collected at field sites. These samples represent a rich resource unique among natural history collections due to NEON's utility for continental- and decadal-scale ecology. NEON's archival samples are available upon request to support research studies and analyses. The comprehensive data, spatial extent and remote sensing technology provided by the NEON project will contribute to a better understanding and more accurate forecasting of how human activities impact ecology and how our society can more effectively address critical ecological questions and issues.

Download activity report

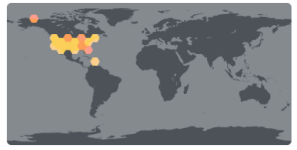
1685 38th St. Ste. 100
Boulder
CO
80305
United States of America
neonscience@battelleecology.org
+1 720-746-4844

Endorsed by: GBIF-US

New data publishers need to be endorsed by

HUMBOLDT

NEON Breeding landbird point counts
This dataset contains quality-controlled breeding landbird survey data collected by the U.S. National Ecological Observatory Network (NEON) across terrestrial sites in the United States as part of a...
Published by National Ecological Observatory Network
Sampling event | 373,518 occurrences | 111 citations



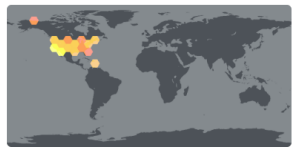
NEON Biorepository Benthic Macroinvertebrate Collection (Taxonomy Subsample)
Taxonomic identifications of bulk benthic macroinvertebrates (exluding chironomids and oligocaetes) collected from wadeable streams, rivers, and lakes. Collected samples are separated into...
Published by National Ecological Observatory Network
Occurrence dataset | 186,227 occurrences | 20 citations

HUMBOLDT

NEON ticks sampled using drag cloths and tick pathogen status
This dataset contains tick survey data collected by the U.S. National Ecological Observatory Network (NEON) across terrestrial sites in the United States as part of a long-term ecological monitoring...
Published by National Ecological Observatory Network
Sampling event | 184,550 occurrences | 49 citations

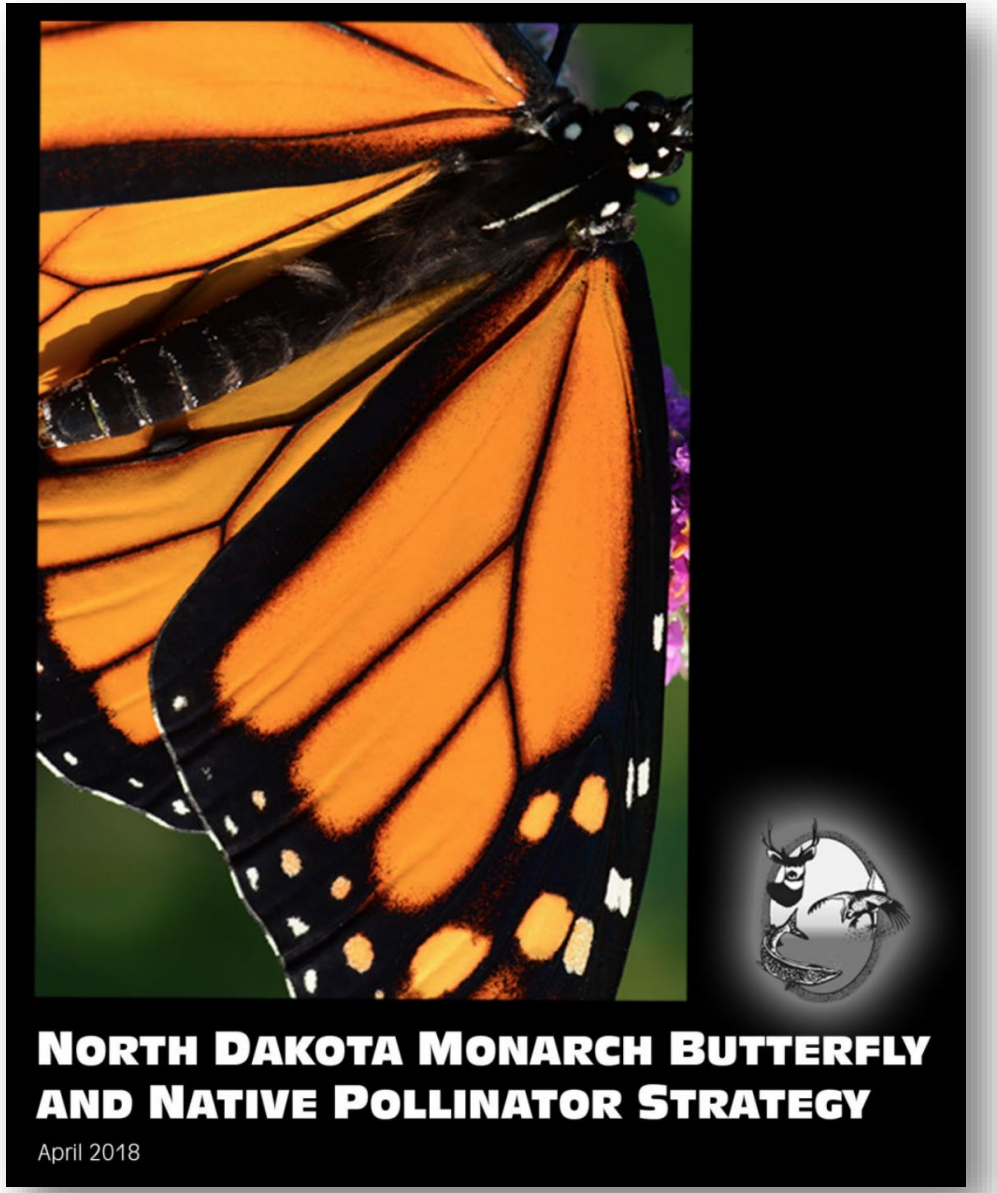


Identified Aquatic Microalgae Collection (Microscope Slides)
Microscope slides for diatom taxonomic reference from periphyton (biofilm) and seston/phytoplankton (water column) aquatic microalgae collections. Sampling 3 sampling bouts per year (spring, summer,...
Published by National Ecological Observatory Network
Occurrence dataset | 122,108 occurrences



NEON Biorepository Carabid Collection (Pinned Vouchers)
Pinned adult Carabid beetles collected from pitfall traps at terrestrial NEON sites. Sampling Catch collected biweekly (every 14 days) during the green season Ten distributed base plots (pre-2023)...
Published by National Ecological Observatory Network
Occurrence dataset | 98,168 occurrences | 116 citations

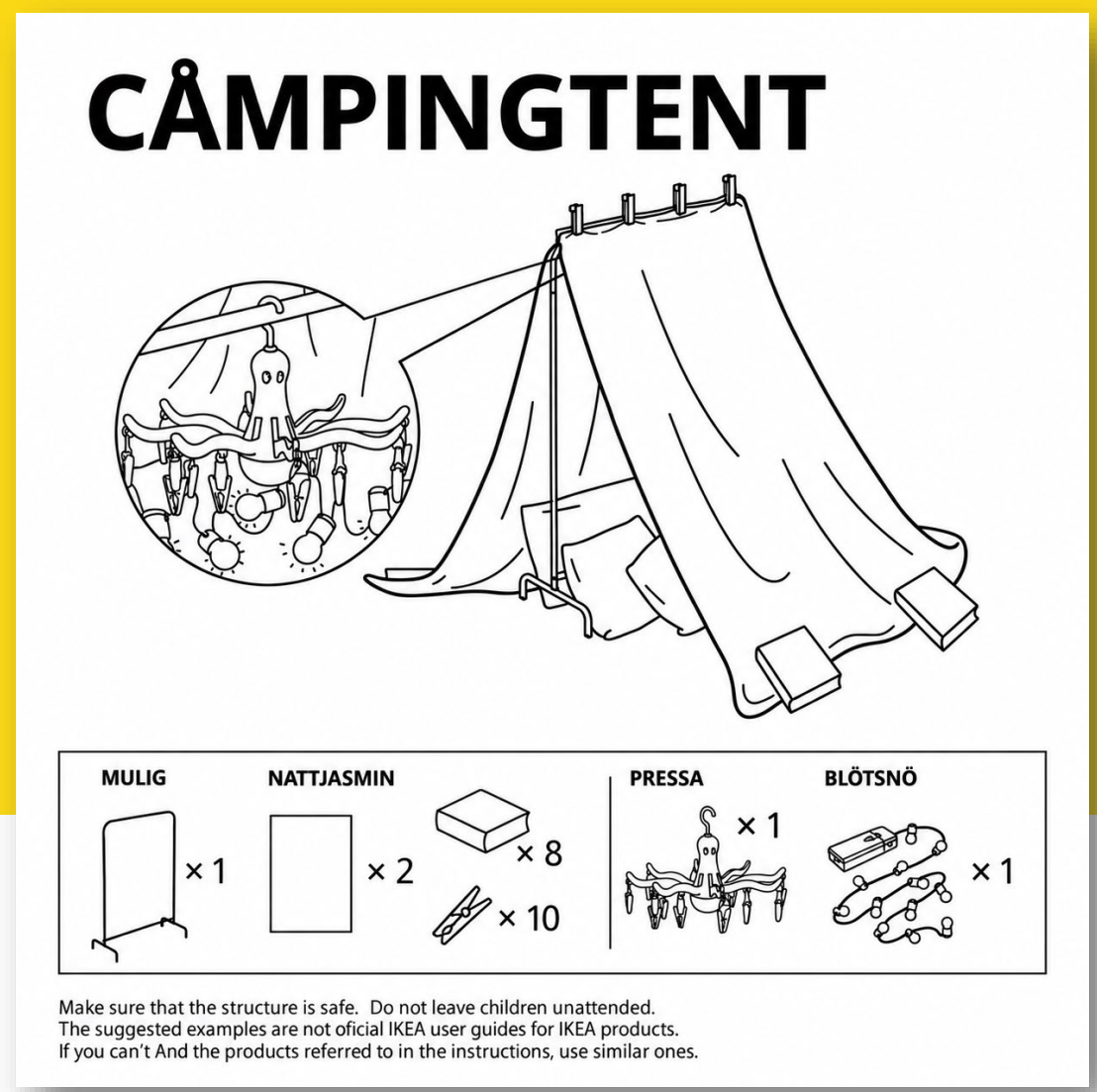




- Statewide butterfly and floral resource survey as part of North Dakota's pollinator conservation efforts
- North Dakota State University, North Dakota Department of Agriculture and the North Dakota Game and Fish Department

IKEA

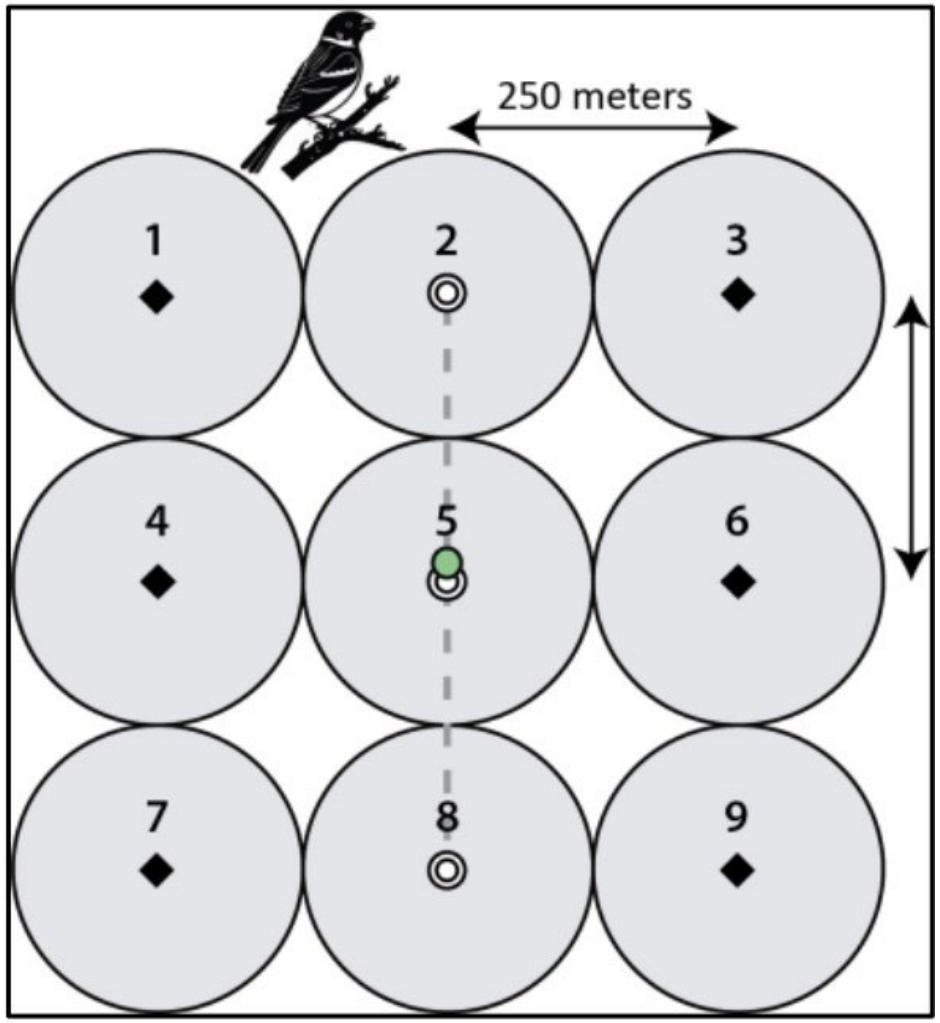




1. dont toss the manual

keep the protocol handy

get as much context as possible



Target: Breeding landbirds active during the day.
Method: Record birds seen or heard during six-minute counts.
Large sites: 5–10 plots with nine survey points each; visited once a year.
Small sites: 2–25 survey points; visited twice a year.
Timing: During the breeding season.
Bird data: Species, how detected, distance, and detection time.
Survey data: Location, date, time, observer, and weather.

you might have to go beyond the protocol

- Humboldt is designed to make datasets self-contained, with as much context as possible
- Every dataset is context dependent
- NEON does not supply information about its domains or sites in the end data, so I had to go get that data

1	domain_id	domain_name	sq_km	decimalLatitude	decimalLongitude	coordinateUncertaintyinMeters	core_terrestrial	core_aquatic	science_questions	gradient
2	D01	Northeast	435750.52	42.89	-74.46	1091799.45	HARV	HOPB	Climate Change; Urbanization	Nitrogen Deposition
3	D02	Mid-Atlantic	351647.28	35.95	-79.65	886557.43	SCBI	POSE	Invasive Species	Nitrogen Deposition; Invaded and Uninv
4	D03	Southeast	402898.58	31	-83.46	1588076.75	OSBS	BARC; SUGG	Forest Management	Forest Management
5	D04	Atlantic Neotropical	35600.06	26.24	-80.79	1948144.11	GUAN	CUPE	Agriculture; Urbanization	Agricultural use
6	D05	Great Lakes	545054.75	44.96	-87.71	1100206.81	UNDE	CRAM	Forest Management	Forest Management
7	D06	Prairie Peninsula	651365.40	41.19	-92.31	939615.08	KONZ	KING	Agriculture	Agricultural use
8	D07	Appalachians & Cumberland Plateau	305961.51	37.62	-84.03	541175.06	ORNL	WALK	Climate Change	Nitrogen deposition
9	D08	Ozarks Complex	682246.11	34.74	-91.01	879868.22	TALL	MAYF	Climate Change	Ecohydrology
10	D09	Northern Plains	866826.73	45.91	-102.72	1273420.14	WOOD	PRPO	Agriculture	Agricultural use
11	D10	Central Plains	453477.61	38.04	-101.67	762604.97	CPER	ARIK	Agriculture; Climate Change	Agricultural use; Dust and Nitrogen Depo

NSF neon Operated by Battelle

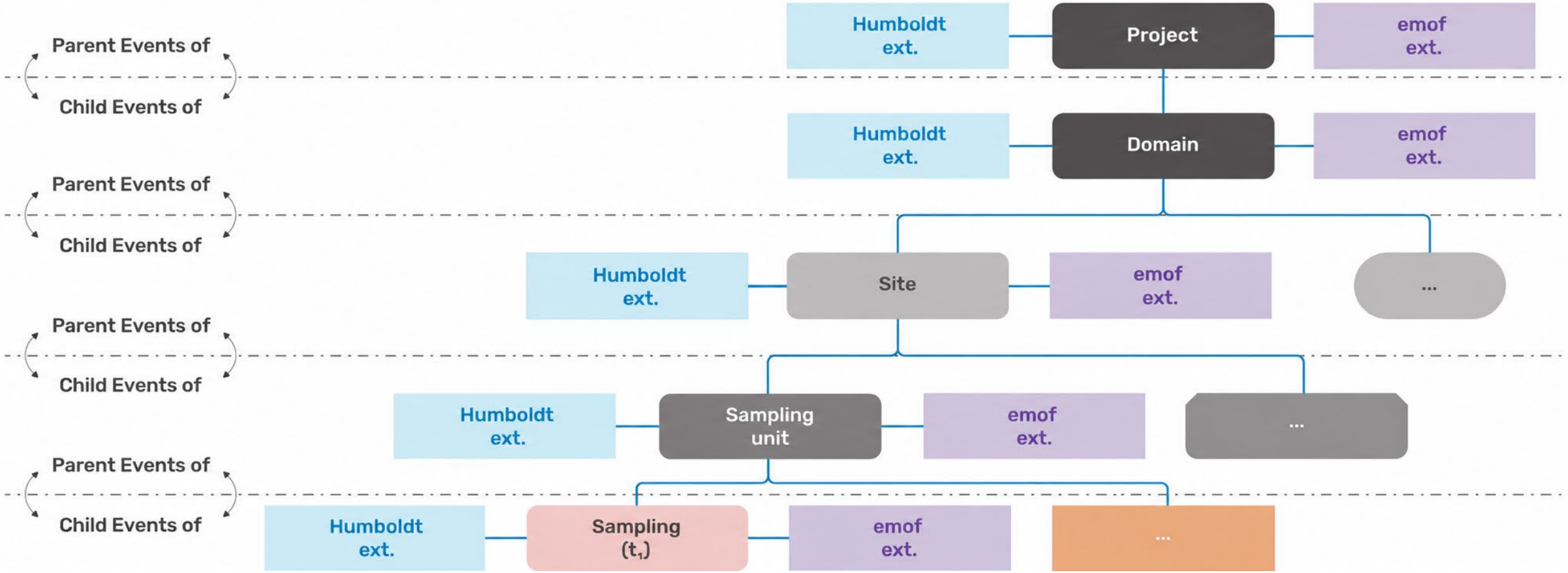


*knolling

2. mise en place your data

consider the events

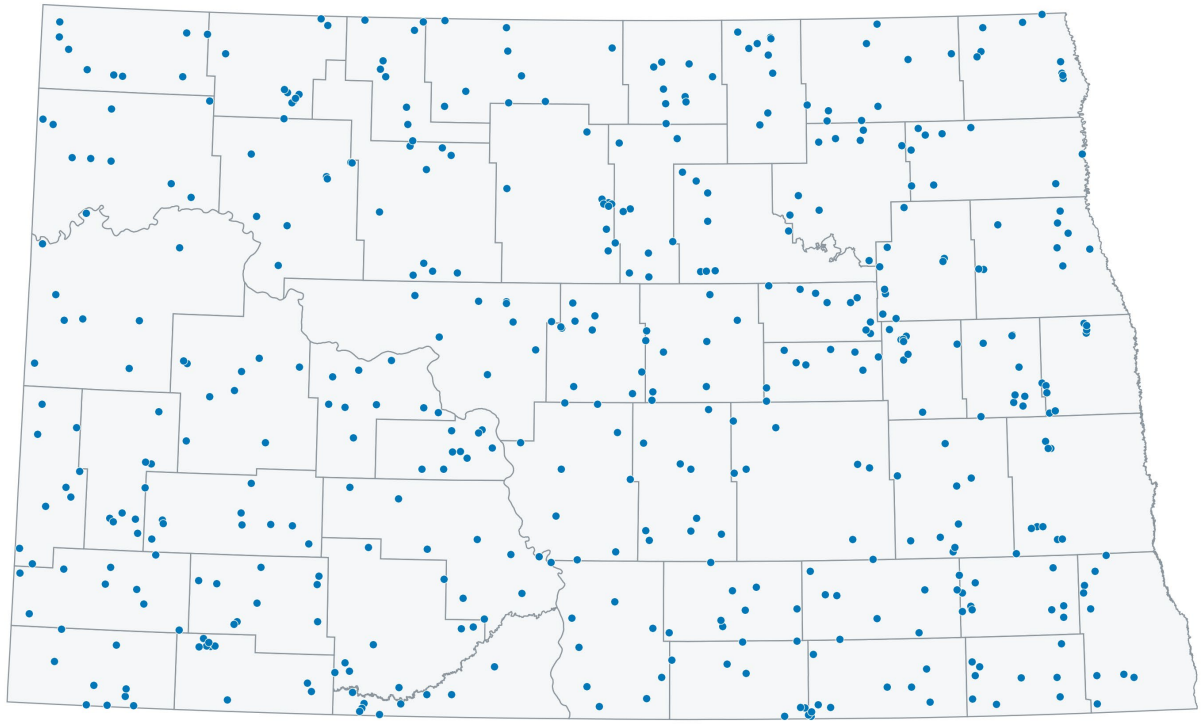
develop a simple, yet inclusive event structure that works for **your data**



you'll likely have to make your own identifiers

North Dakota butterfly survey sites

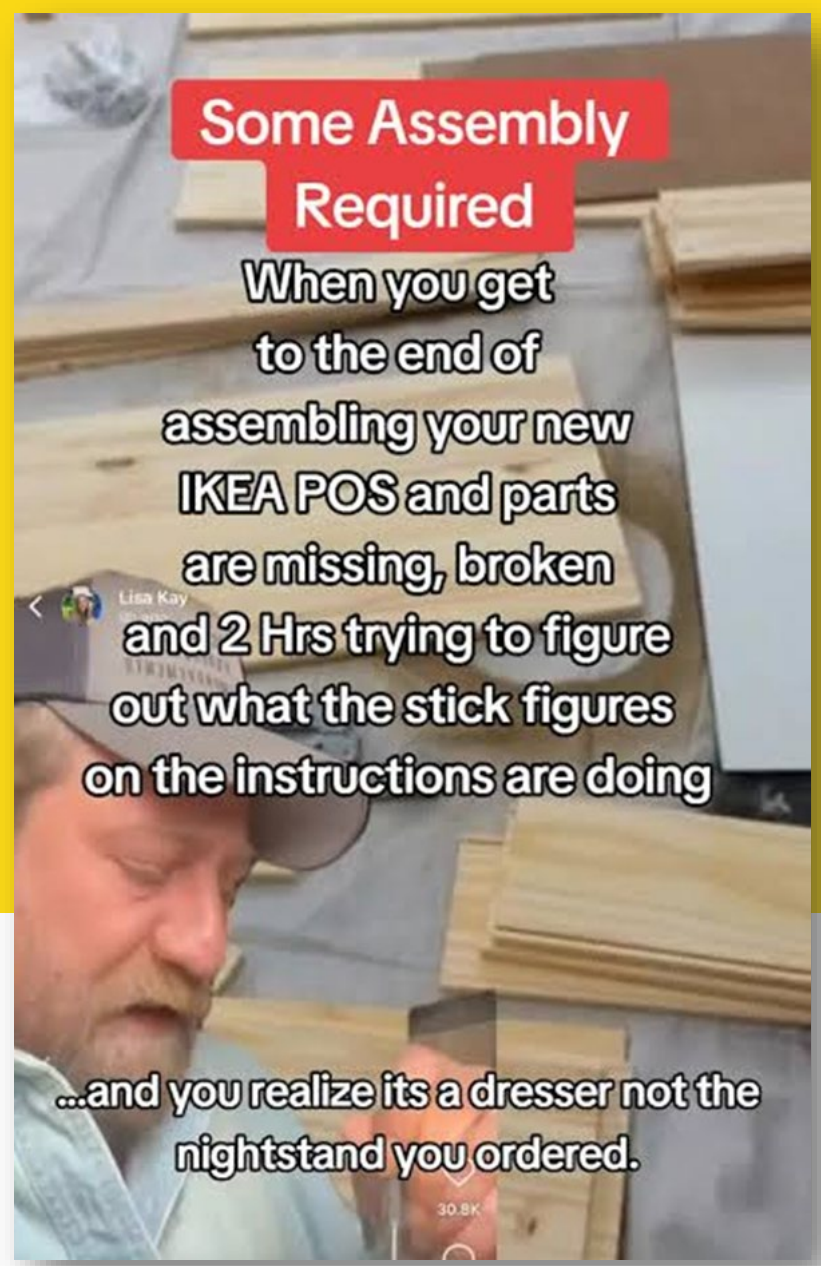
474 mapped sites across 53 counties



Raw IDs	Edited IDs
--SurveyID-- LTDS-A0198 VES-A0198	--Site Visit ID— A0198 A0198

- IDs often contain data (i.e., site+date)
- Can use UUIDs
- Pros and cons to each

Check uniqueness, unmatched keys, and join row counts.



3. some re^assembly required

theres no way you did it right the first time, sorry

map highest level to lowest

- Map Data
  - **Map Event Core & Humboldt Data**
    - Project Event
    - Domain Events
    - Site Events
    - Plot Events
    - Plot Visit Events
    - Merge Event Tables
  - Map Occurrence Data
  - Map Resource Relationship Data
  - Map Extended Measurement or Fact



https://biokic.github.io/NEONTickstoHumboldt

NEON Ticks & Tick Pathogens

https://biokic.github.io/NEONTickstoHumboldt

https://www.gbif.org/dataset/12315bb8-8ab3-446a-b5a4-2be93aade242

NEON Birds

https://biokic.github.io/NEONBirdstoHumboldt/

https://www.gbif.org/dataset/516eb7c0-3586-4dfc-bc18-c1fc1e765b7f



SYCA – Domain 14, Desert Southwest

Operated by Battelle

720.746.4844 | neonscience@battelleecology.org | NEONscience.org